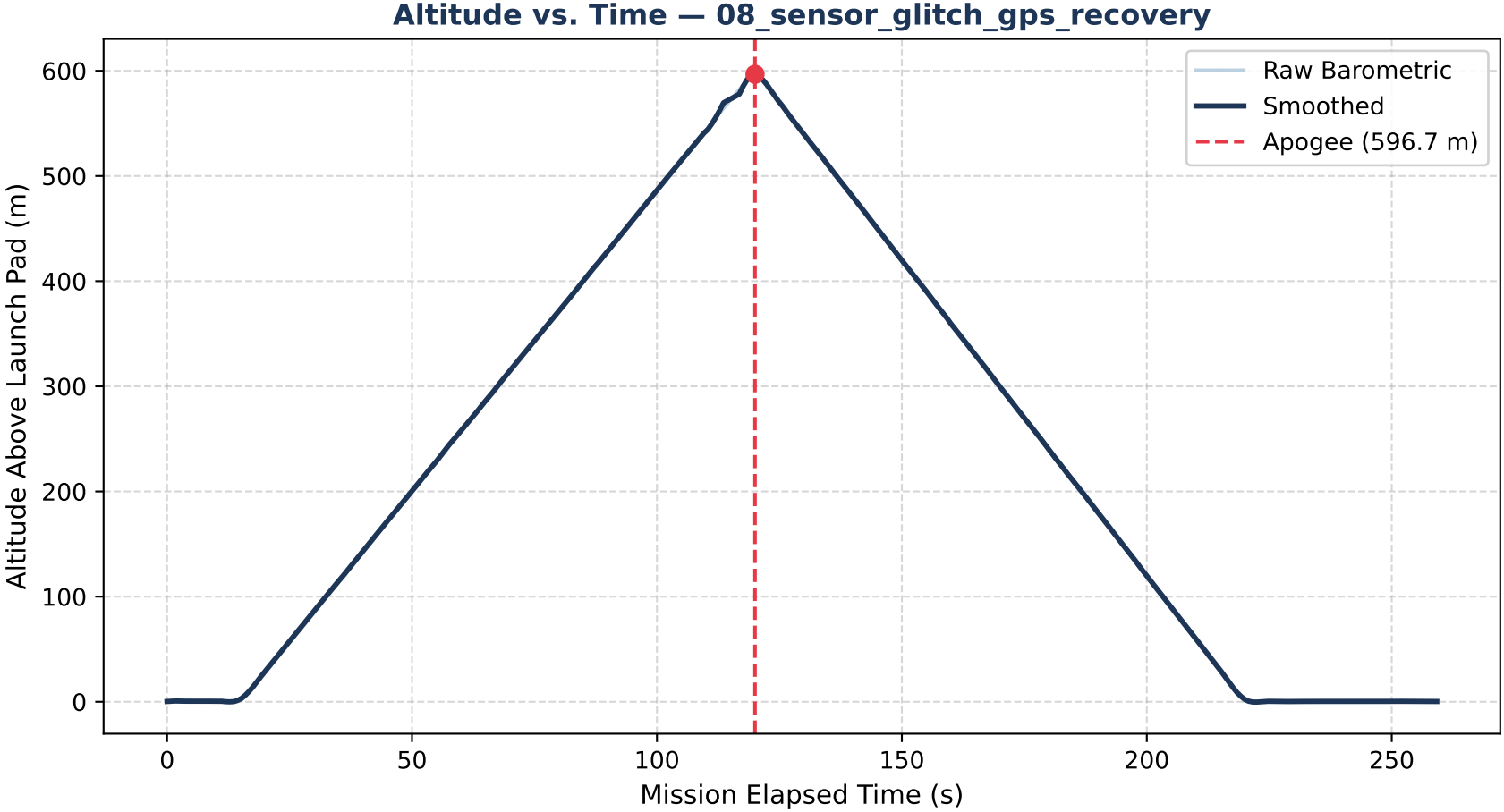
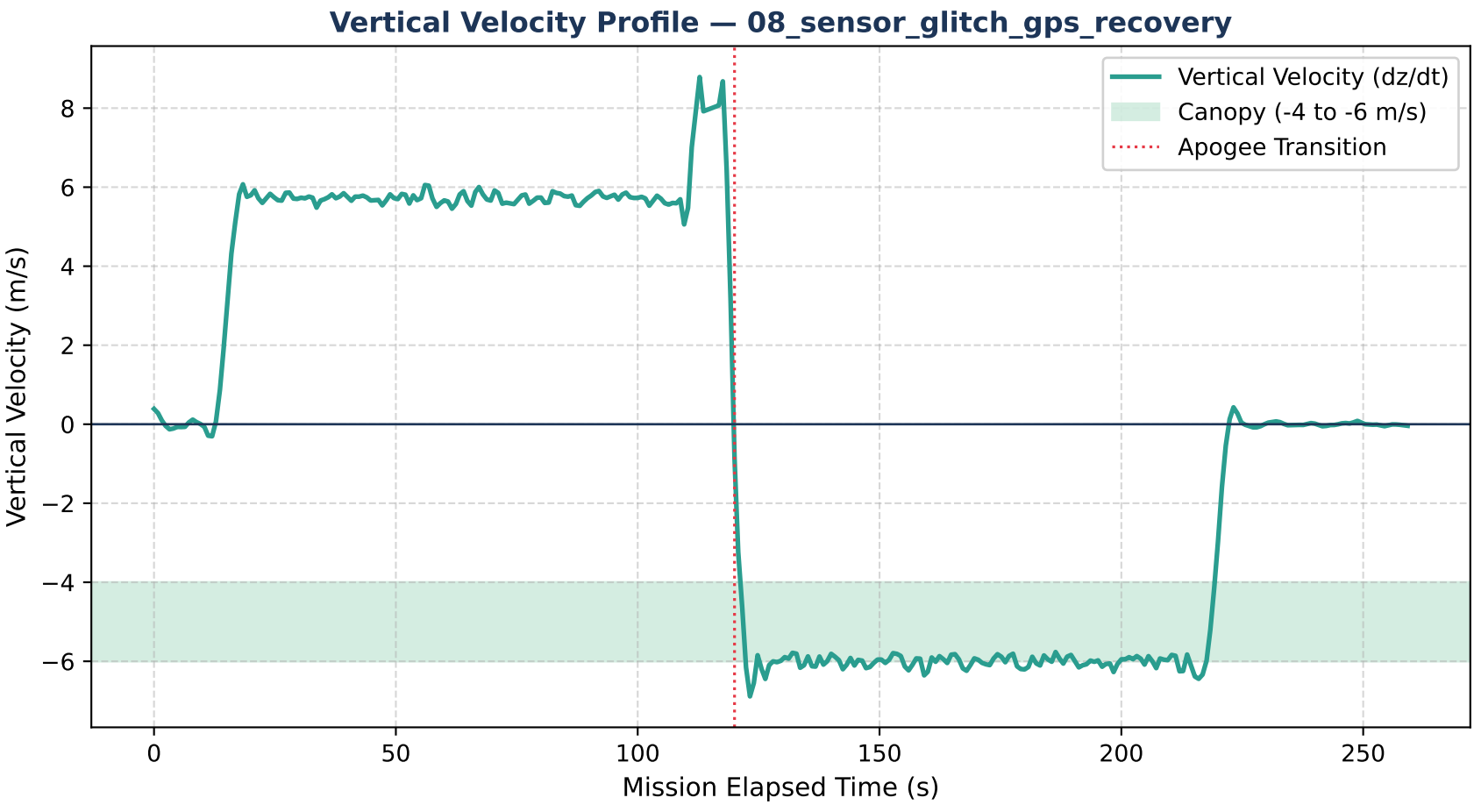
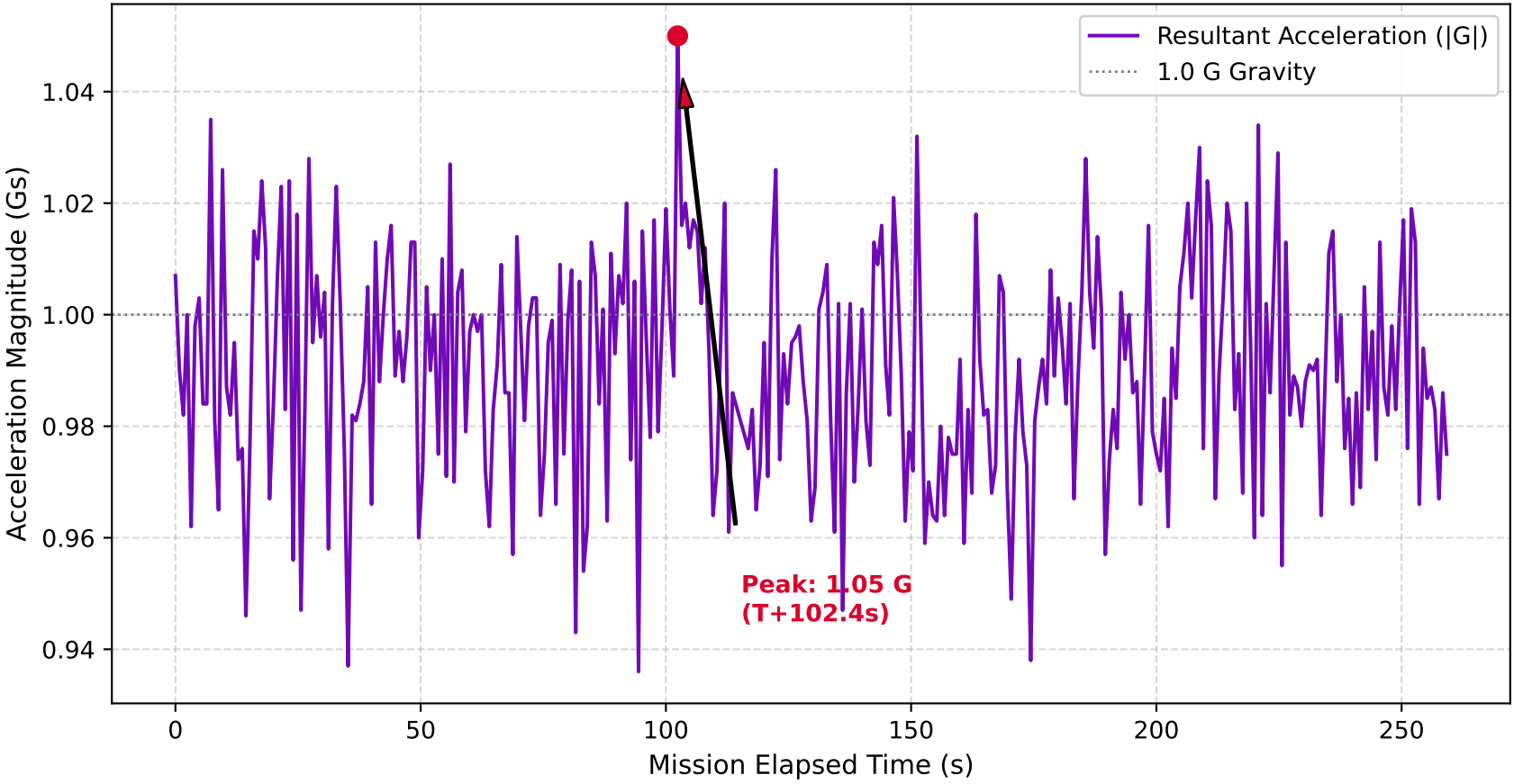
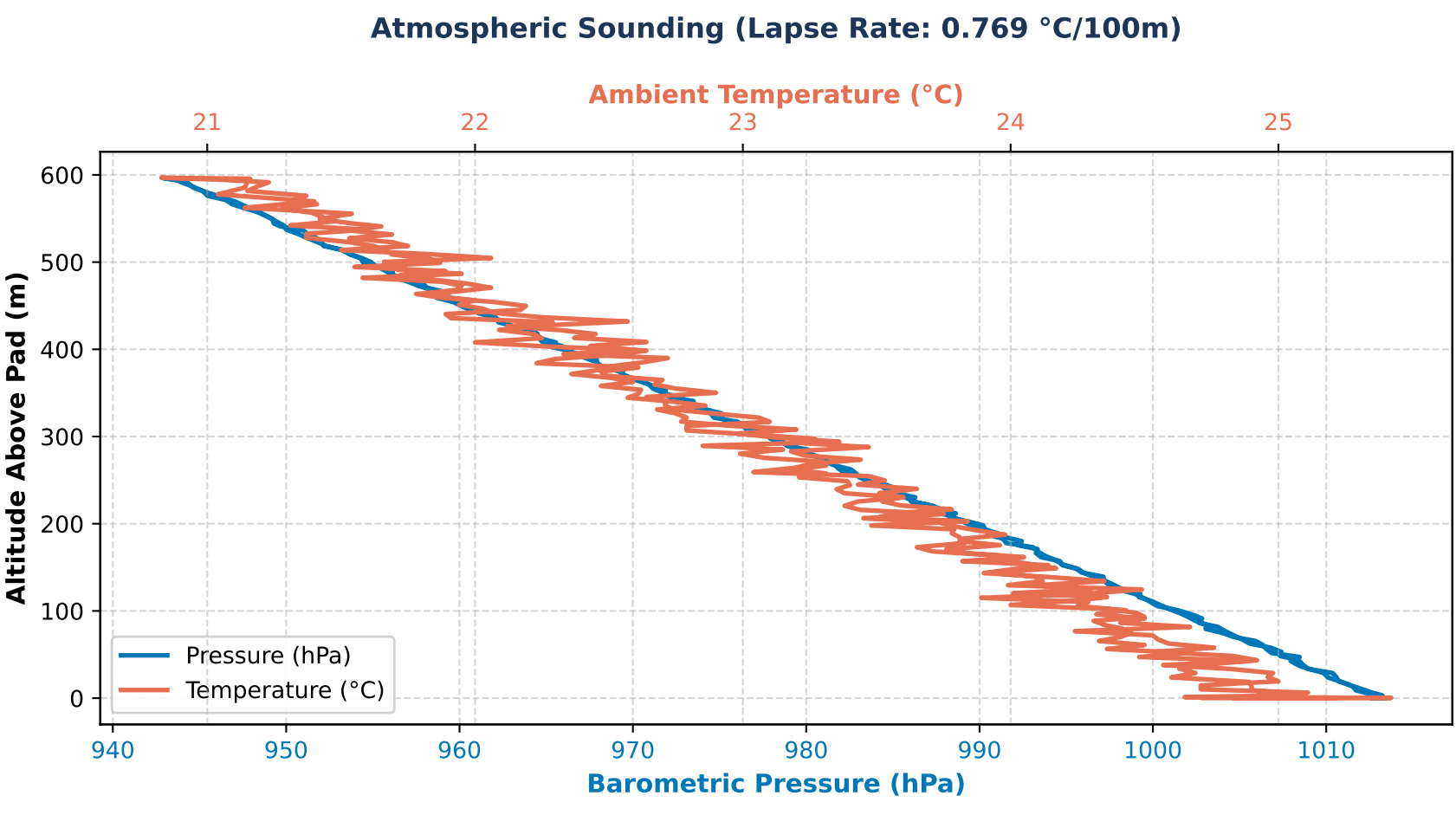
Cognitive CanSat - Mission Summary: 08_sensor_glitch_gps_recovery

Flight Dynamics Parameter	Calibrated Value
Mission Name	08_sensor_glitch_gps_recovery
Apogee (m)	596.72
Time to Apogee (s)	120.02
Max Ascent Velocity (m/s)	8.79
Max Descent Velocity (m/s)	-6.89
Max G-Shock (Gs)	1.05
Touchdown Shock (Gs)	1.03
Mission Duration (s)	259.19
Min Temperature (°C)	20.83
Max Temperature (°C)	25.42
Min Pressure (hPa)	942.82
Max Pressure (hPa)	1013.73
Lapse Rate (°C/100m)	0.769
Telemetry Frames	322
Landing Lat	22.575138
Landing Lon	88.368553


Altitude vs. Time — 08_sensor_glitch_gps_recovery
Raw Barometric
Smoothed
Apogee (596.7 m)
Altitude Above Launch Pad (m)
0
100
200
300
400
500
600
0
50
100
150
200
250
Mission Elapsed Time (s)


Vertical Velocity Profile — 08_sensor_glitch_gps_recovery
Vertical Velocity (dz/dt)
Canopy (-4 to -6 m/s)
Apogee Transition
Vertical Velocity (m/s)
8
6
4
2
0
−2
−4
−6
0
50
100
150
200
250
Mission Elapsed Time (s)


G-Shock & Acceleration — 08_sensor_glitch_gps_recovery
Resultant Acceleration (|G|)
1.0 G Gravity
Peak: 1.05 G
(T+102.4s)
Acceleration Magnitude (Gs)
Mission Elapsed Time (s)
1.04
1.02
1.00
0.98
0.96
0.94
0
50
100
150
200
250


Atmospheric Sounding (Lapse Rate: 0.769 °C/100m)
Ambient Temperature (°C)
21
22
23
24
25
Altitude Above Pad (m)
600
500
400
300
200
100
0
Pressure (hPa)
Temperature (°C)
940
950
960
970
980
990
1000
1010
Barometric Pressure (hPa)

Cognitive CanSat Mission Dynamics: 08_sensor_glitch_gps_recovery

